Δ DELTA FILINGS

Corporate Governance Intelligence Platform

Functional Requirements Document

Version 2.0 — April 2026

Companion to: Delta Filings Business Requirements Document

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1. Introduction

1.1 Purpose

This Functional Requirements Document (FRD) decomposes the business requirements from the Delta Filings BRD into detailed functional specifications. Each feature block captures the purpose, actors, flow, alternate flows and acceptance criteria necessary for implementation and verification.

1.2 Audience

- Engineering — implementation reference.
- QA — test-case derivation.
- Product — scope and acceptance alignment.
- Design — flow-by-flow UX contract.

1.3 Conventions

- IDs follow the pattern FR-<module>-<sequence>.
- Priority uses MoSCoW (Must / Should / Could).
- 'System' refers to the Delta Filings platform as a whole.
- All timestamps are in IST unless stated otherwise.

1.4 Traceability to BRD

FRD Module	BRD Requirements
Dashboard (FR-D)	BR-001 — BR-005 (Filings)
Announcements (FR-A)	BR-001 — BR-005
DIN Search (FR-N)	BR-010 — BR-014
Compare (FR-C)	BR-012, BR-014
Compliance (FR-M)	BR-020 — BR-024
Watchlist (FR-W)	BR-030 — BR-033
Accounts (FR-U)	BR-040 — BR-043

2. Dashboard Module (FR-D)

FR-D-01 — Display latest filings across user's watchlists

Attribute	Value
ID	FR-D-01
Priority	Must
Primary Actor(s)	Authenticated user
Purpose	Give the user an at-a-glance view of the most recent disclosures for companies they track.
Preconditions	User is authenticated; at least one watchlist exists.
Postconditions	User sees up to 50 filings, newest first.

Main Flow

1. User navigates to /dashboard.
2. System loads watchlists and associated companies for the user.
3. System queries the filings service for the latest 50 filings across those companies.
4. System renders a card list with company, category, timestamp and AI summary.
5. User can click a card to open the filing detail page.

Alternate / Error Flows

- No watchlists yet — system shows an empty-state CTA to create a watchlist.
- Filings service unavailable — system shows a cached snapshot with a 'stale data' badge.

Acceptance Criteria

- Dashboard renders within 2.5s on a 4G connection (p95).
- Dashboard respects watchlist membership changes within 60s.
- No filings older than the most recent 50 are shown.

FR-D-02 — Show AI-generated summary with sentiment tag

Attribute	Value
ID	FR-D-02

Attribute	Value
Priority	Must
Primary Actor(s)	Authenticated user
Purpose	Reduce time-to-insight by pre-summarising each filing.
Preconditions	A filing has been ingested and summary pipeline has completed.
Postconditions	User sees 1–3 sentence summary with a Positive/Neutral/Negative badge.

Main Flow

6. System reads the cached summary from the filings row.
7. System renders the summary text with a sentiment badge.
8. System displays a 'machine-generated' label for transparency.

Alternate / Error Flows

- Summary not yet generated — show 'Summary pending' placeholder and enqueue a summary job.
- Summary generation failed — show raw first 200 chars of filing body with 'AI unavailable' note.

Acceptance Criteria

- All summaries carry a visible 'machine-generated' label.
- Sentiment badge uses an accessible colour contrast (≥ 4.5:1).

3. Announcements Module (FR-A)

FR-A-01 — Ingest BSE and NSE filings every 60 seconds

Attribute	Value
ID	FR-A-01
Priority	Must
Primary Actor(s)	Ingestion worker (system actor)
Purpose	Keep the filings corpus within 15 minutes of source publication.
Preconditions	Worker has network access to BSE and NSE public feeds; Redis available.

Attribute	Value
Postconditions	New filings persisted to Postgres with de-duplication; summary jobs enqueued.

Main Flow

9. Scheduler triggers the scraper every 60 seconds during market hours.
10. Scraper fetches the latest N announcements per exchange.
11. Scraper normalises the payload into the canonical Filing schema.
12. System checks Redis for a dedup key (sha256 of source URL + timestamp).
13. If not seen, system inserts into Postgres and enqueues a summarisation job.

Alternate / Error Flows

- Source returns 5xx — back off with exponential delay up to 5 minutes.
- Source layout changes — raise parse-error alert and continue with last known structure.

Acceptance Criteria

- Lag between source publication and Delta ingestion is < 15 minutes at p95.
- Zero duplicate filings persisted over any 24-hour sample window.

FR-A-02 — Filing detail page

Attribute	Value
ID	FR-A-02
Priority	Must
Primary Actor(s)	Any user
Purpose	Give users a canonical view of a single filing.
Preconditions	Filing row exists in Postgres.
Postconditions	User sees structured filing with source link and related filings.

Main Flow

14. User opens /announcements/{filingId}.
15. System fetches filing, summary and related filings (same company, last 5).
16. Page renders metadata, summary, embedded PDF viewer and related list.

Alternate / Error Flows

- Filing not found — show 404 with a 'back to announcements' CTA.
- PDF unavailable — show link to original source with explanatory note.

Acceptance Criteria

- Filing detail page fully loads within 3s at p95.
- Source URL always matches original BSE/NSE publication URL.

4. DIN Search Module (FR-N)

FR-N-01 — Look up director by 8-digit DIN

Attribute	Value
ID	FR-N-01
Priority	Must
Primary Actor(s)	Any user
Purpose	Return structured director details quickly and reliably.
Preconditions	User has an 8-digit DIN.
Postconditions	User sees the director's name, DOB (year only), and company associations.

Main Flow

17. User types DIN into /din search box.
18. System validates DIN is exactly 8 digits.
19. System checks Redis cache for a fresh (≤ 24h) response.
20. If cache miss, system calls MCA scraper or returns curated snapshot.
21. System renders director profile with association list.

Alternate / Error Flows

- Invalid DIN — inline field-level error, no request sent.
- MCA unavailable — fall back to curated snapshot with 'curated snapshot' badge.

Acceptance Criteria

- DIN lookup p95 latency < 800ms on cache hits.
- Curated-snapshot badge visible whenever data did not come from live MCA.

FR-N-02 — Redis caching for DIN responses

Attribute	Value
ID	FR-N-02
Priority	Must
Primary Actor(s)	System
Purpose	Absorb load and protect the MCA source from rate limits.
Preconditions	Redis reachable.
Postconditions	Director payload cached for 24h keyed by DIN.

Main Flow

22. System writes successful MCA payload to Redis with TTL of 24h.
23. Subsequent lookups within TTL return from cache.
24. Cache invalidation runs daily at 02:00 IST for curated snapshots.

Alternate / Error Flows

- Redis unreachable — system logs a warning and serves direct MCA/curated data.

Acceptance Criteria

- Cache hit ratio ≥ 85% in production over a 30-day sample.

5. Compare Module (FR-C)

FR-C-01 — Compare up to three directors

Attribute	Value
ID	FR-C-01
Priority	Must
Primary Actor(s)	Authenticated user
Purpose	Expose relationships between directors at a glance.
Preconditions	User is authenticated.
Postconditions	User sees a side-by-side grid of up to three directors.

Attribute	Value

Main Flow

25. User navigates to /compare.
26. User adds directors by DIN or name search.
27. System renders a three-column grid with name, DOB (year), associations, flags.
28. System highlights common directorships across columns.

Alternate / Error Flows

- User tries to add a fourth — system disables the 'add' control with tooltip.
- Director not found — inline error on the add control.

Acceptance Criteria

- Common directorships visually distinguished (badge + colour).
- Grid remains readable at 1280px viewport width.

FR-C-02 — Shareable read-only comparison link

Attribute	Value
ID	FR-C-02
Priority	Should
Primary Actor(s)	Authenticated user
Purpose	Let users share a compare view with collaborators without login.
Preconditions	User has created a comparison.
Postconditions	Public read-only URL is generated.

Main Flow

29. User clicks 'Share'.
30. System generates a UUIDv4 token stored server-side with a 30-day expiry.
31. System returns a URL of the form /share/compare/<token>.

Alternate / Error Flows

- User revokes the link — token deleted and URL returns 404.

Acceptance Criteria

- Shared URL is read-only; no edit affordances rendered.
- Token expiry enforced server-side, not just in the UI.

6. Compliance Module (FR-M)

FR-M-01 — Show obligations relevant to a company

Attribute	Value
ID	FR-M-01
Priority	Must
Primary Actor(s)	Compliance officer (authenticated user)
Purpose	Surface the applicable regulatory obligations for a given company and its sector.
Preconditions	User is authenticated and has selected a company.
Postconditions	User sees a filtered obligations list with due dates.

Main Flow

32. User navigates to /compliance?companyId=…
33. System fetches obligations where applicability matches the company's sector and listing.
34. System renders the list grouped by statute (LODR, Companies Act, SEBI circulars).

Alternate / Error Flows

- Unknown sector — system shows all obligations marked 'review applicability manually'.

Acceptance Criteria

- Obligation list load < 1s at p95.
- Each obligation links to its source statute / circular.

FR-M-02 — Create a compliance run with evidence capture

Attribute	Value
ID	FR-M-02
Priority	Must
Primary Actor(s)	Compliance officer
Purpose	Record the execution of a compliance obligation for audit.
Preconditions	Obligation selected; user has 'compliance-write' permission.
Postconditions	Run row persisted with evidence attachments and timestamp.

Main Flow

35. User clicks 'Create run' on an obligation.
36. User fills due date, assignee and notes.
37. User uploads one or more evidence files (PDF, PNG).
38. System stores files in object storage and persists the run.
39. System generates an audit log entry.

Alternate / Error Flows

- Upload fails — show per-file error; run remains draft until resolved.
- File > 25 MB — rejected with inline error.

Acceptance Criteria

- Run is immutable once marked 'complete' (evidence hash recorded).
- Audit log entries never lose their chain when a run is updated.

FR-M-03 — Export compliance run as PDF

Attribute	Value
ID	FR-M-03
Priority	Should
Primary Actor(s)	Compliance officer, external auditor
Purpose	Produce an audit-ready artefact.

Attribute	Value
Preconditions	Run exists with at least one piece of evidence.
Postconditions	A PDF is generated and delivered to the user.

Main Flow

40. User clicks 'Export PDF' on a run.
41. System renders a server-side PDF with metadata, evidence list, and hashes.
42. System returns a signed download URL (expires in 10 minutes).

Alternate / Error Flows

- PDF generation fails — show retry CTA; log to observability stack.

Acceptance Criteria

- PDF includes SHA-256 hash of each evidence file.
- PDF export latency < 10s at p95 for runs with up to 20 evidence files.

7. Watchlist and Alerts Module (FR-W)

FR-W-01 — Create and manage watchlists

Attribute	Value
ID	FR-W-01
Priority	Must
Primary Actor(s)	Authenticated user
Purpose	Let users group companies they care about.
Preconditions	User is authenticated.
Postconditions	Watchlist persisted under the user's account.

Main Flow

43. User opens /watchlists.
44. User clicks 'New watchlist' and provides a name.
45. User adds companies by ticker or name search.

Alternate / Error Flows

- Duplicate name — system shows inline validation error.

Acceptance Criteria

- User can create up to 20 watchlists on the Free tier.
- Watchlist membership changes reflect on the dashboard within 60s.

FR-W-02 — Email alerts on new filings

Attribute	Value
ID	FR-W-02
Priority	Must
Primary Actor(s)	System
Purpose	Notify the user of material filings in their watchlists.
Preconditions	User has verified email; watchlist has ≥ 1 company.
Postconditions	Alert email dispatched.

Main Flow

46. Background job polls for new filings within user's watchlists.
47. System applies user's cadence preference (instant / daily / weekly).
48. System dispatches an email via transactional provider with unsubscribe link.

Alternate / Error Flows

- Email bounces — system marks email 'bounced', pauses future alerts until user action.

Acceptance Criteria

- Instant alerts delivered within 2 minutes of ingestion at p95.
- Every alert includes a working one-click unsubscribe.

8. Account Management (FR-U)

FR-U-01 — Register and sign in via Supabase

Attribute	Value
ID	FR-U-01
Priority	Must
Primary Actor(s)	Visitor
Purpose	Authenticate users securely without building credential storage.
Preconditions	Visitor has a valid email or Google account.
Postconditions	Supabase issues a JWT; session cookie set on the Next.js domain.

Main Flow

49. Visitor clicks 'Sign in' and chooses email or Google.
50. Supabase handles the flow and returns a signed JWT.
51. Next.js middleware validates the JWT and sets a secure httpOnly cookie.

Alternate / Error Flows

- OAuth cancelled — return to /login with a neutral state.
- Supabase unavailable — show status-page link and graceful error.

Acceptance Criteria

- All sessions use httpOnly, Secure, SameSite=Lax cookies.
- Sign-in-to-dashboard round-trip < 2s at p95.

FR-U-02 — Delete account within 30 days

Attribute	Value
ID	FR-U-02
Priority	Must
Primary Actor(s)	Authenticated user
Purpose	Honour the user's right to deletion.

Attribute	Value
Preconditions	User is authenticated.
Postconditions	User data purged from Postgres, Redis and object storage.

Main Flow

52. User opens /account/delete.
53. User confirms by typing their email.
54. System enqueues a delete job with a 7-day grace period.
55. Job removes rows and blobs; audit log entry retained.

Alternate / Error Flows

- User cancels within grace period — job cancelled; no data removed.

Acceptance Criteria

- All PII fields hard-deleted within 30 days of initial request.
- Audit log entry remains and carries the pseudonymous hash only.

9. Non-Functional Cross-Reference

Non-Functional Area	Applicable FRs
Performance	FR-D-01, FR-A-02, FR-N-01, FR-M-03, FR-U-01
Availability	FR-A-01, FR-W-02
Security	FR-U-01, FR-U-02, FR-M-02
Accessibility	All user-facing FR-D / FR-A / FR-N / FR-C / FR-M / FR-W
Observability	FR-A-01, FR-W-02, FR-M-03